

Our Environment & Natural Calamities (Part- 2)

Introduction to Pollution

Everything around us is the component of our environment. The components of our environment affect on our health and day to day life. We get disease in the polluted environment and a good environment makes us healthy. The polluting elements in our environment are called pollutants. In this chapter, we will study about the pollutants in our environment and method by which we can eliminate these pollutants.

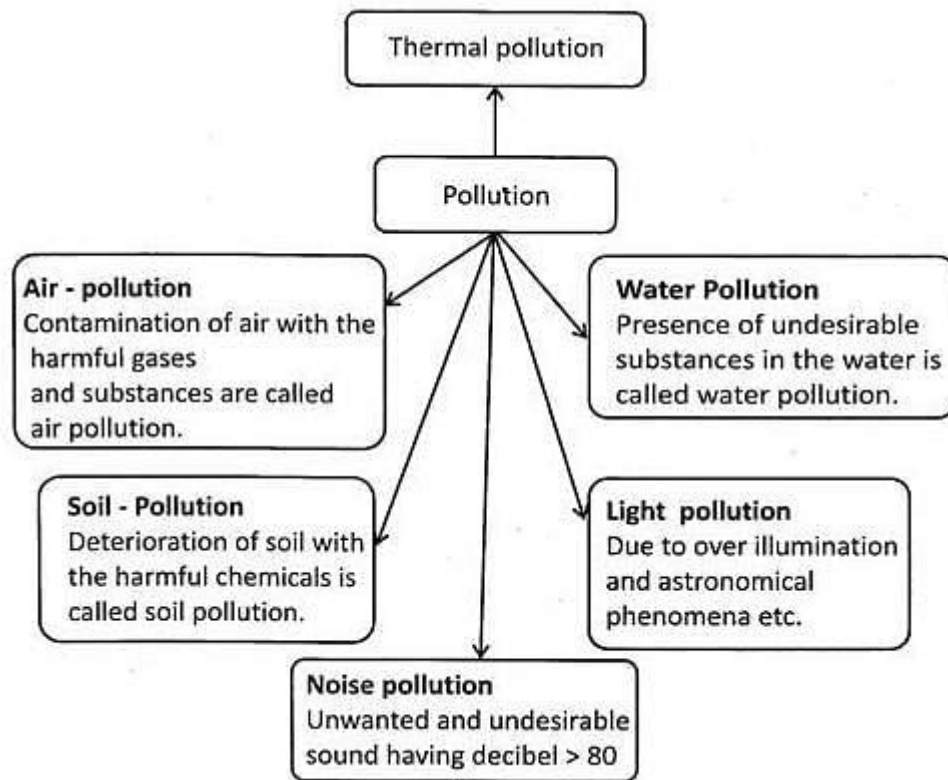
Pollution



Fig: Air Pollution

Human activities lead to the production of waste materials that had deteriorated our environment and make them unsuitable for the life. Unwanted and undesirable substances present in air, water and soil are called as pollutant. For example. Sulphur dioxide, Nitrogen oxides, Lead oxides. Hydrocarbons, Fly ashes, Smoke, Dust particles etc.

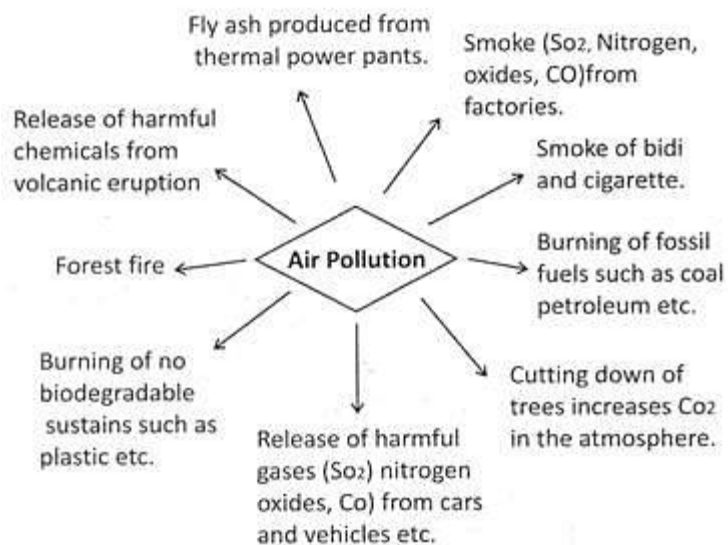
On the basis of the contaminated source, pollution can be broadly classified into the following types:

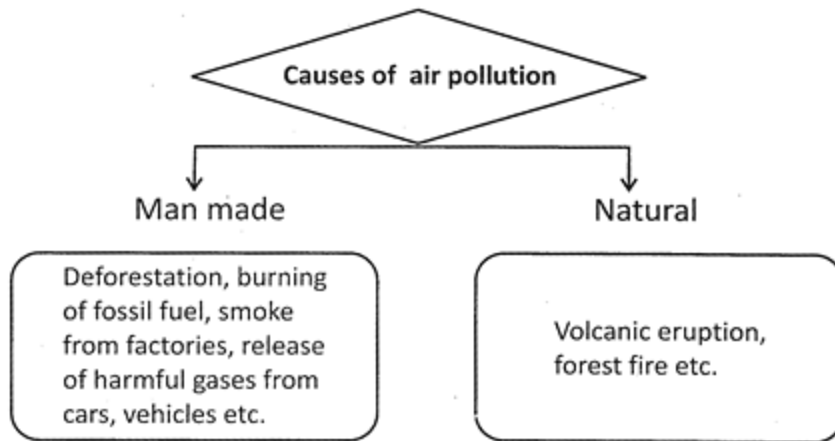


Air Pollution

Overpopulation, industrialization, modern life style and advancement in science technology has lead to the thoughtless use of natural resources, which in turn spoil air for other activities. This is called air pollution.

Look at the following picture of the causes of air pollution



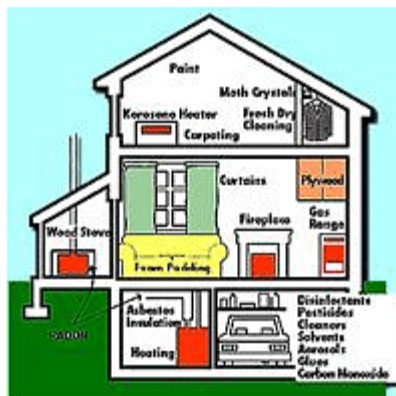


Deforestation is one of the big causes of air pollution. Trees are the natural air purifier. Deforestation increase the concentration of carbon dioxide in the air and thereby pollutes it.



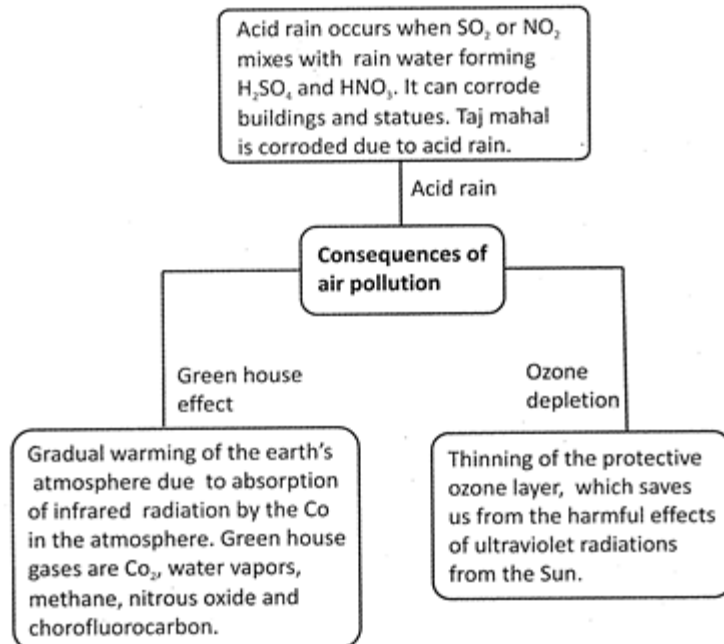
Fig: Natural cause of pollution

Eruption of ash from the volcano causes air pollution. Concentration of carbon dioxide, hydrogen chloride and sulfur dioxide increases due to the eruption from volcano



Indoor pollutants which are responsible for the air pollution, given in the picture above. We can minimize the pollution by using nonpolluting elements instead of the given elements.

Consequences of Air Pollution

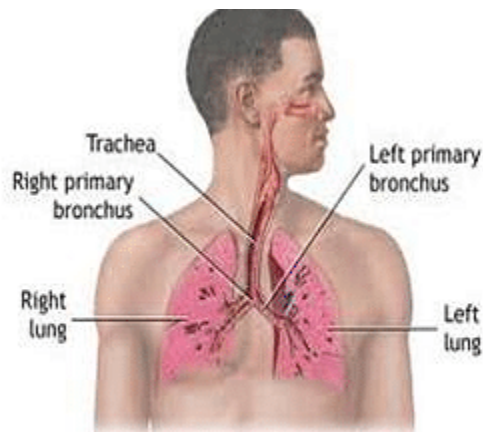


Air Pollution Causes Diseases



Pollution causes itching in eyes

Many respiratory problems Asthma, Bronchitis occur due to inhalation of poisonous gases.

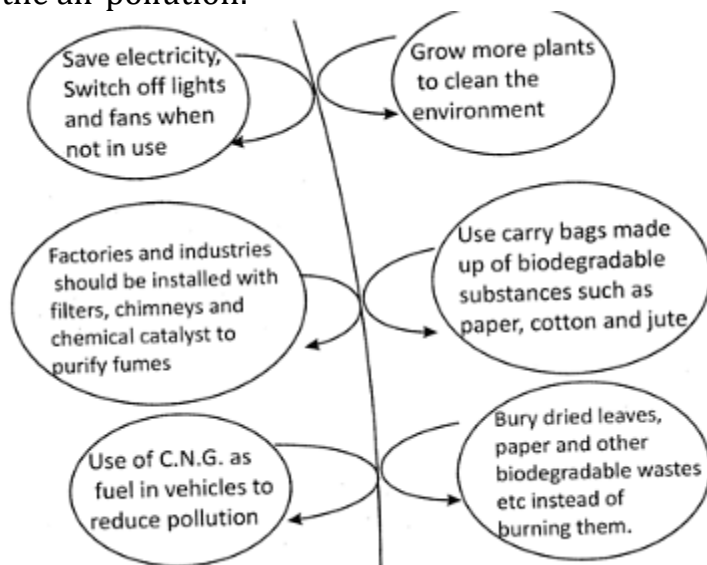


Respiratory problems

- Pollution is injurious to plant or other animal also.
- Polluted air can damage lungs.
- It reduces visibility because of the presence of smoke, dust and vapors.
- Pollution can damage our nervous system and digestive system.
- Pollution can cause cancer and neurological disorder.

How to Minimize Air Pollution

Look at the following chart to understand the method by which, we can minimize the air pollution:



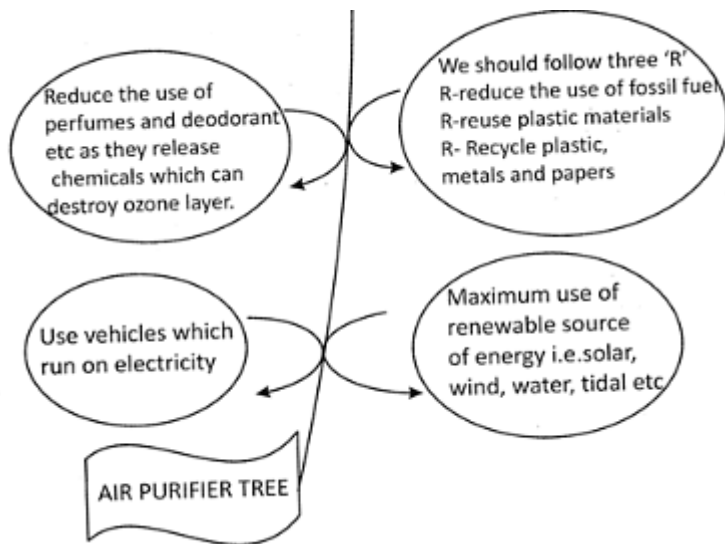


Fig: An Eco-Friendly Car

Use electric vehicle to reduce release of harmful gases in the air



Fig: An eco friendly bus

Commonly Asked Questions

If there would be no ozone layer, then which harmful products would not be

screened by the atmosphere?

- (a) Smog
- (b) UV rays
- (c) IR rays
- (d) All of these
- (e) None of these

Answer: (b)

Explanation: Ozone layer in the stratosphere layer of the atmosphere filters the ray of UV to come on the surface of the earth. Due to air pollution ozone layer is depleting, hence allowing UV rays to enter into the earth's atmosphere. Therefore, option (b) is correct and rest of the options is incorrect.

Water Pollution

Water is an essential component of all forms of life. It is odorless, colorless and transparent liquid. It has the capacity to dissolve maximum substances in it and hence is used as a universal solvent. But now a days due to human activities, over population, industrialization and unawareness, the smell, odor and unique quality of water is being contaminated, which is called water pollution.



Fig: Water Pollution

Polluted water causes disease

Water borne diseases can spread through contaminated water. Contaminated water contains pathogenic microorganisms.

How to minimize water pollution

Look at the following chart to understand the method by which we can minimize water pollution:

Commonly Asked Questions:

1. Which one of the following is a factor of water pollution?

- (a) Factory's wastes
- (b) Pesticides, insecticides, weedicides
- (c) Untreated sewage
- (d) All of these
- (e) None of these

Answer: (d)

Explanation

Factory's wastes. Pesticides, insecticides, weedicides and untreated sewage all are the factors of water pollution. Therefore, option (d) is correct and rest of the options is incorrect.

2. What may be the consequences of water pollution?

- (a) Killing of aquatic animals
- (b) Disturbance in the ecological balance
- (c) Disturbance in the food chain
- (d) All of these
- (e) None of these

Answer: (d)

Explanation

All are the possible consequences of water pollution that have been given in the option A, B, and C. Therefore, option (d) is correct and rest of the options is incorrect.

Soil Pollution

Excessive, unorganized agriculture and no judicious use of chemical fertilizers, pesticides, weedicides, herbicides and insecticides etc contaminate the soil which is called soil pollution. Polluted soil leads to the growth of polluted plants and this toxicity finally enters the human/ animal's body through food chain.

Causes of Soil Pollution

- Seepage from a landfill.
- Discharge of industrial waste into the soil.
- Percolation of contaminated water into the soil.
- Rupture of underground storage tanks.
- Excess application of pesticides, herbicides or fertilizer.
- Solid waste seepage.
- Deforestation and Soil erosion.

Polluted Soil Causes Disease

How can we maintain the quality of soil?

Look at the following chart to understand the method by which, we can maintain the quality of soil

Commonly Asked Questions

1. Find the statement which is true about soil pollution?

- (a) Brain and nerve can be damaged due to soil pollution
- (b) Malaria is caused by the protozoa that live in soil
- (c) Metals, pesticides, solvents, lead and oil spills are soil contaminants
- (d) Deforestation leads to soil pollution
- (e) All of these

Answer: (e)

Explanation

Soil pollution causes many diseases. Hence, all the statements in the options are true. Therefore, option (e) is correct and rest of the options is incorrect.

2. Consider the following statements:

Statement 1: Climate of a region is affected due to soil pollution.

Statement 2: Climate of a region is not affected due to soil pollution.

Which one of the following is correct about the above statements?

- (a) Statement 1 is true and 2 is false
- (b) Statement 1 is false and 2 is true
- (c) Both statements are true
- (d) Both statements are false
- (e) None of these

Answer; (a)

Explanation

Soil pollution affects vegetation which in turn affects climate. Therefore, option (a) is correct and rest of the options is incorrect.

Noise Pollution

It is the loud unwanted and undesirable sound. Sound at 80 decibel or more gives unpleasant experience and regarded as noise pollution. 90% of noise pollution is due to the noise produced by the motor vehicles etc.

Look at the following picture of the causes of noise pollution:



Fig: Cause of noise pollution

Classification of Sound in Decibel (db)

Sounds	Decibel (db)
Normal talking	40 db
Whisper	20 db
Aircraft	120 db
Atomic explosion	200 db

Intensity of sound greater than 80 db causes noise pollution.

Consequences of Noise Pollution

- **Emotional or psychological effects** - irritability, anxiety and stress. Lack of concentration and mental fatigue are significant health effects of noise.
- It has been observed that the performance of school children is poor in comprehension tasks when schools are situated in busy areas of a city and suffer from noise pollution - disturbance.
- **Interferes with normal auditory communication**, it may mask auditory warning signals and hence increases the rate of accidents especially in industries.
- The effects can range in severity from being extremely annoying to being extremely painful and hazardous **Lowers workers efficiency and productivity** and higher accident rates on the job.

Noise Causes Disease



Fig: Ill effects of noise pollution

Our ears are sensible organs which work properly on a specific range of sound.

The brain receives the incoming signal from the ears and analyze it. If the intensity of sound increases, it can damage the nervous system and ultimately may lead to hearing loss. The intensity of sound, which is greater than the sensible capacity of our ears is called loud and unpleasant sound. The effect of unpleasant sound on our head is just like inserting nails in the sensitive head.

Commonly Asked Questions

Arwin is playing violin and his sister Tina is listening music from the loudspeaker. Which one of the two children is creating more noise pollution as compare to other?

- (a) Arwin
- (b) Tina
- (c) Both Arwin and Tina
- (d) Neither Arwin nor Tina
- (e) None of these

Answer: (b)

Explanation

Sound of violin is pleasant but sound of a loudspeaker is always noise because it is at high volume. Therefore, option (b) is correct and rest of the options is incorrect.

Consider the following statements:

Statement 1: A fire engine produces noise pollution.

Statement 2: Leaves falling off a trees produces noise pollution.

Which one of the following is correct about the above statements?

- (a) Statement 1 is true and 2 is false
- (b) Statement 1 is false and 2 is true
- (c) Both 1 and 2 are false
- (d) Both 1 and 2 are true
- (e) None of these

Answer: (a)

Explanation

Falling of leaves from the trees produces sound of very low decibel that doesn't come under pollution. Therefore, option (a) is correct and rest of the options is incorrect.

You Must Know:

A plastic milk jug takes million years to decompose.

More than 100 active pesticides ingredients are suspected of causing cancer.

Untreated sewage is a significant risk to human health and wealth.

Nitrogen is essential for crops but too much will leak into underground and rivers causing water pollution.